1. Odd-n trace accounting: who owns the factor 2 on $2E$

2026-11-07. A follow-up to the subgroup-trap study (PR #3646), which derived, without measuring, that for odd n the twisted/standard yield ratio on $2E$ targets is 2^{n+1}, not 2^n. The protocol was frozen before any study data at `3516d8a885`. The run is `reports/run-20261107T1/`. Every statement is an unverified assertion of the producing session [pcc:unverified].

1.1. Setup

$E : y^2 + xy = x^3 + ax^2 + b$ over $\mathbb{F}_{2^N}$, $N = dn$, $q = 2^d$, with $\deg b = N$. For $P \neq O$, $P \in 2E$ iff $\mathrm{Tr}(x_P) = \mathrm{Tr}(a)$. A line $u\mathbb{F}_q$ is trace-homogeneous iff $\mathrm{Tr}_{N/d}(u) = 0$, since $\mathrm{Tr}_N(ut) = \mathrm{Tr}_d\big(t\mathrm{Tr}_{N/d}(u)\big)$. The plain $\mathbb{F}_q$ has $\mathrm{Tr}_{N/d}(1) = n \bmod 2$, so it is trace-mixed for odd n.

Arms: `std` ($\mathbb{F}_q$); `tz` (its $2E$ part); `tw` (all-lift line: $\mathrm{Tr}_{N/d}(u) = \mathrm{Tr}_{N/d}\big(\sqrt{b}/u\big) = 0$); `twh` (the half of `tw` with $\mathrm{Tr}_d(t) = 0$); `hz` (a trace-homogeneous line that is not all-lift, so it has the standard size); and `mix` (a generic trace-mixed line). Cells: $n = 3$ ($d = 3, 4, 5$), $n = 5$ ($d = 3, 4, 5$), and $n = 4$ ($d = 3, 4$) as the even control. Three curves with $\mathrm{Tr}(a) = 0$ per cell, plus one with $\mathrm{Tr}(a) = 1$. Targets: every point when $N \leq 15$, otherwise 2,000 per class, in the classes $2E$, off $2E$, and odd order.

Exact coset identity. $\sum_{R \in 2E} N(R) = \big((m_0 + m_1)^n + (m_0 - m_1)^n\big)/2$, where m_0 and m_1 count base points in and off $2E$. Among bases of size m, the $2E$ mean is largest, $2m^n/\#E$, exactly for single-coset bases. Write $\kappa = \text{mean } N \cdot \#E/m^n$ (1 for a uniform base, 2 for a single-coset base on its coset) and $\eta = \text{mean } N/c$ (relations per trial per column).

1.2. Results

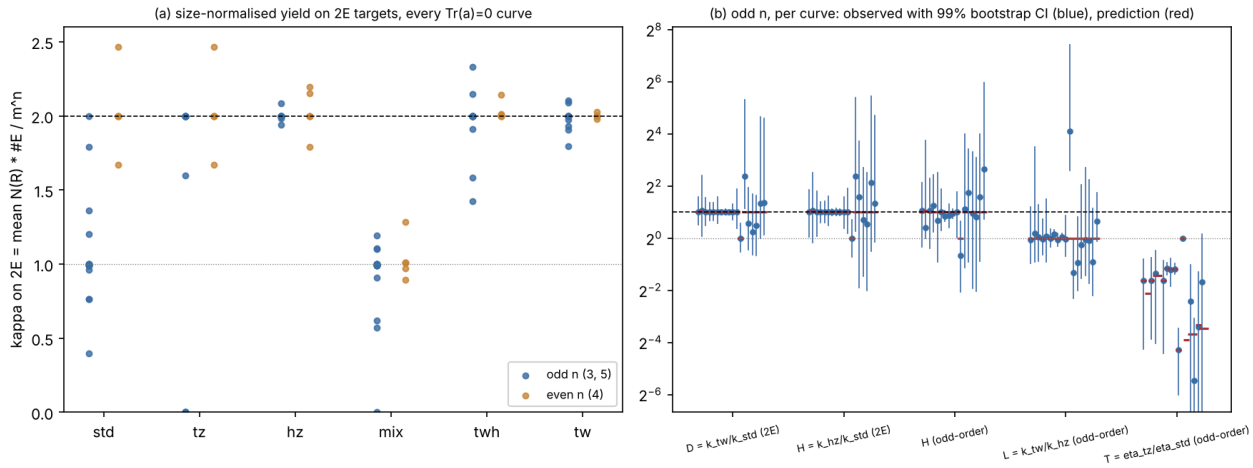


Figure 1: (a) κ on $2E$ targets for every arm and every $\mathrm{Tr}(a) = 0$ curve. Trace-homogeneous arms sit at 2; `std` sits at 1 for odd n and at 2 for $n = 4$. (b) Per-curve ratios for odd n, with 99 % bootstrap intervals and the composition-exact predictions.

Exact checks (P8, PASS).

- The trace criterion agrees with $[\#E/2]R = O$ on every target, with 0 mismatches.
- MITM counts agree with brute force on 2,293,041 (arm, target) pairs.
- The coset identity holds exactly on all 120 census arm-units.
- `tw` has exactly $2(q-1)$ points, and `tw`, `twh`, `hz` and `tz` lie in $2E$.
- For odd n, all 469,256 off-$2E$ counts of `tz`, `tw`, `twh` and `hz` are 0.

Registered verdicts.

- P2 (all targets) PASS. P4 ($n = 4$) PASS. P7 ($\mathrm{Tr}(a) = 1$) PASS. P8 PASS.
- P1 FAIL. The pooled $\kappa_{\mathrm{tw}}/\kappa_{\mathrm{std}}$ on $2E$ is 1.96 for odd n (inside the band [1.7, 2.3]) and 0.996 for even n. Only 11 of 18 per-curve intervals exceed 1.25, where 17 were needed.
- P3, P5 and P6 FAIL, and the interpretation rule returns **inconclusive**. Every failure comes from $n = 5$ units, apart from one $n = 3$ curve (`tz` on a 65-point odd subgroup). The causes:
  - extremely clustered counts: 97 to 100 % zeros for the small arms;
  - zero sample means, which make the registered geometric mean of per-curve ratios 0, ∞ or NaN;
  - one $q = 8$ curve whose 6-point $\mathbb{F}_q$ base is trace-homogeneous by chance;

- two tiny odd subgroups, of 65 and 257 points.

Post hoc pooled analysis (not registered; units pooled as the mean of κ over curves, with a bootstrap over targets within curves):

group, class	$H = K_{\mathrm{hz}}/K_{\mathrm{std}}$	$L = K_{\mathrm{tw}}/K_{\mathrm{hz}}$	$D = K_{\mathrm{tw}}/K_{\mathrm{std}}$
odd n, all 18, $2E$	2.08 [1.62, 2.72]	0.90 [0.71, 1.11]	1.86 [1.62, 2.16]
odd n, `std` mixed (17), odd order	2.11 [1.55, 2.80]	1.00 [0.80, 1.26]	2.10 [1.70, 2.65]
$n = 4$, 6, $2E$	1.00 [0.80, 1.26]	0.99 [0.84, 1.17]	0.99 [0.84, 1.19]

On the $n = 4$ control, the trace-mixed line has $K_{\mathrm{mix}}/K_{\mathrm{std}} = 0.51$ [0.40, 0.66], the mirror image of the odd-n effect. For `tz`, $\eta_{\mathrm{tz}}/\eta_{\mathrm{std}}$ on $2E$ is 0.91 [0.70, 1.13] of its composition prediction, which is about 2^{2-n}. Restricting $\mathbb{F}_q$ to its trace-zero half therefore loses per column.

1.3. Reading

- The extra 2 for odd n is a real per-trial gain, at equal base size, of any trace-homogeneous base over the trace-mixed $\mathbb{F}_q$.
- An ordinary index calculus gets it by using a trace-homogeneous line `hz` of the standard size, with the same system shape.
- So it belongs to the baseline, not to the twisted line. Crediting 2^{n+1} to the twisted line is accounting.
- For the Oakley lane ($n = 5$, large q), on cryptographic targets:
  - against plain $\mathbb{F}_q$: $2^{n+1} = 64\times$ yield per trial, $32\times$ in trials;
  - against `hz`: $2^n = 32\times$ yield, $16\times$ in trials.
- This reading is post hoc. The registered rule was inconclusive, and the solver cost of `hz` was not measured.

1.4. Not claimed

No ECDLP cost, leaderboard, benchmark or rho row. Execution was local exact counting; the AGENTS.md §9a isolab-eks confirmatory run is open. The KB was unavailable (no `research-kb` binary, MCP not connected), so the lead atlas and `rg` were used instead.